

Statement of Work: Boiler Repair, Cleaning, and Calibration

1. Summary

This Statement of Work (SOW) outlines the requirements for boiler repairs, cleaning, and control calibration at The Defense Logistics Agency at Susquehanna. The work is divided into four (4) Contract Line Item Numbers (CLINs).

- CLIN 0001: Replacement of one (1) ruptured waterwall tube and membrane section within Nebraska Boiler C8 (NB#4529, Boiler #2)
 - Nebraska, model number: NS-E-64-250, serial number: CP-4088
- CLIN 0002: Clearing of obstructions from the boiler header and associated tubes of Boiler #2.
 - Nebraska, model number: NS-E-64-250, serial number: CP-4088
- CLIN 0003: Chemical cleaning of the two (2) remaining boilers to remove scale and ensure proper flow.
 - Boiler #1- Nebraska model number NS-E-64-250; serial number CP-4089
 - Boiler #3- Nebraska model number NS-49-250; serial number CP-4090
- CLIN 0004: Calibration of the boiler controls for all three (3) boilers.

The contractor shall provide all labor, materials, equipment, and supervision necessary to safely and effectively complete all CLINs as detailed below.

2. Scope of Work

CLIN 0001: Ruptured Waterwall Tube Replacement - Boiler #2

The Contractor shall perform the following tasks to replace the ruptured tube section in Boiler #2.

- 2.1. Site Assessment and Preparation:
 - 2.1.1. Conduct a pre-job site visit to visually inspect the damaged tube and surrounding area.
 - 2.1.2. Verify the exact location of the rupture on the waterwall tube.
 - 2.1.3. Identify and document any obstructions or access challenges.

- 2.1.4. Review and adhere to all plant safety regulations, including LOTO procedures and confined space entry requirements.
- 2.1.5. Coordinate with boiler plant operators to ensure the boiler is properly shut down, cooled, and isolated.
- 2.1.6. Prepare the work area by erecting scaffolding, removing insulation as needed, and establishing proper ventilation.
- 2.2. Removal of Damaged Tube and Membrane:
 - 2.2.1. Precisely measure and document the dimensions of the existing tube section to be replaced.
 - 2.2.2. Carefully cut and remove the damaged section of the waterwall tube and adjacent membrane.
 - 2.2.3. Grind and prepare the existing tube ends for welding to ensure a clean, beveled surface.
 - 2.2.4. Properly dispose of the removed tube and membrane section per regulations.
- 2.3. Fabrication and Installation of Replacement Tube and Membrane:
 - 2.3.1. Fabricate replacement tube and membrane sections matching the original specifications and meeting ASME Boiler and Pressure Vessel Code requirements.
 - 2.3.2. Position and weld the replacement components into place using qualified welders and approved procedures.
 - 2.3.3. Perform visual inspection of all welds.
- 2.4. Post-Weld Treatment and Inspection:
 - 2.4.1. Perform post-weld heat treatment (PWHT) on the welded joints, if required.
 - 2.4.2. Provide all Non-Destructive Examination (NDE) reports to the plant for review and approval.
- 2.5. Hydrostatic Testing:
 - 2.5.1. Coordinate with plant operators to hydrostatically test the repair at the required pressure for a minimum of 30 minutes.

- 2.5.2. If leaks are detected, repair the welds and re-perform the test until no leaks are present.
 - 2.5.3. Document the final hydrostatic test results.
 - 2.6. Cleanup and Demobilization:
 - 2.6.1. Remove all scaffolding, tools, and debris from the work area and restore any removed insulation.
 - 2.6.2. Demobilize from the site upon completion of CLIN 001 work.
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CLIN 0002: Obstruction Clearing - Boiler #2

The Contractor shall inspect and clear the boiler header and associated tubes of Boiler #2 to ensure there are no blockages.

- 2.7. Inspection and Clearing:
 - 2.7.1. Conduct a borescope inspection of the boiler header and accessible tubes to identify any obstructions, scale, or debris.
 - 2.7.2. If obstructions are found, remove them using approved mechanical methods (e.g., brushing, snaking) or a localized chemical flush.
 - 2.7.3. All chemical usage must be pre-approved by the plant and handled in accordance with environmental regulations.
 - 2.7.4. Perform a post-cleaning inspection to verify that all obstructions have been cleared.
 - 2.7.5. Document the findings of the inspection and the methods used for clearing.
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CLIN 0003: Chemical Cleaning of Remaining Two Boilers

The Contractor shall perform a full chemical cleaning on the two (2) boilers not addressed in CLIN 001 to remove internal deposits and restore optimal flow.

- 2.8. Chemical Cleaning Procedure:
 - 2.8.1. Coordinate with plant operators to isolate each boiler individually for cleaning.

- 2.8.2. Perform a pre-cleaning flush with water to remove loose sediment.
 - 2.8.3. Introduce and circulate an approved, inhibited chemical cleaning agent to dissolve scale and mineral deposits.
 - 2.8.4. Monitor key parameters during cleaning, including chemical concentration, temperature, and iron levels.
 - 2.8.5. Once cleaning is complete, neutralize and passivate all internal surfaces to prevent flash rusting.
 - 2.8.6. Properly containerize and dispose of all cleaning chemicals and waste materials per regulations.
 - 2.8.7. Conduct a final flush and perform a visual inspection to confirm the effectiveness of the cleaning.
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CLIN 0004: Boiler Controls Calibration

The Contractor shall inspect, test, and calibrate the primary control and safety devices for all three (3) boilers.

- 2.9. Control System Calibration:
 - 2.9.1. Inspect all primary boiler control and safety devices, including but not limited to, pressure sensors, temperature sensors, level switches, fuel-air ratio controllers, and flame scanners.
 - 2.9.2. Calibrate all analog and digital control instruments against certified, traceable standards.
 - 2.9.3. Verify and document the operational setpoints and trip points of all safety interlocks, including low-water cutoff, high-pressure limit, and high/low gas pressure switches.
 - 2.9.4. Perform functional tests to ensure the control system responds correctly to simulated plant conditions and that the boiler executes a safe shutdown upon a trip condition.
 - 2.9.5. Document all "as-found" and "as-left" calibration data for each device.
 - 2.9.6. Provide a final calibration certificate for each boiler upon completion.
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3. Materials

The Contractor shall provide all materials necessary to complete the work for all CLINs, including but not limited to:

- Replacement tube and membrane sections (SA-192 Carbon Steel, A36 Carbon Steel, or equivalent)
- Welding consumables, grinding wheels, and cutting discs
- Scaffolding and insulation materials
- Hydrostatic testing equipment and PPE
- Borescope or other inspection equipment
- All required chemical cleaning agents, neutralizing solutions, and passivation chemicals
- Certified calibration instruments and testing equipment for boiler controls

4. Personnel Qualifications

The Contractor shall employ qualified personnel for all tasks, including:

- Certified welders with experience in boiler tubes.
- Supervisors with experience in boiler repair and chemical cleaning.
- Personnel trained in the safe handling and disposal of industrial chemicals.
- Certified instrumentation and control technicians with experience in boiler control systems.

5. Safety

The Contractor shall be responsible for ensuring the safety of all personnel and shall comply with all applicable OSHA regulations and plant safety procedures. A pre-job safety briefing shall be conducted for all work.

6. Schedule

The Contractor shall complete all work within the stated period of performance and shall provide a detailed schedule of activities for all four CLINs.

7. Deliverables

The Contractor shall provide the following deliverables:

- Detailed schedule of activities
- Material certifications
- Welding Procedure Specifications (WPS) and welder qualification records
- NDE reports
- Hydrostatic test report for CLIN 001
- Inspection and obstruction clearing report for CLIN 002
- Chemical cleaning and waste disposal report for CLIN 003
- Boiler controls calibration report with "as-found" and "as-left" data for CLIN 004

8. Warranty

The Contractor shall warrant all workmanship and materials for a period of 1 year from the date of completion.

9. Owner Furnished Information/Assistance

- 10.1. Plant operators will assist with the testing of all repairs and controls.
- 10.2. Plant operators will assist with filling and draining the boiler for hydro testing.
- 10.3. The plant will provide potable water for hydro testing and flushing.